

# Data Analysis Prelim Exam

Department of Statistics, University of Chicago

Assigned Monday Sept 15 2025 at 12pm (noon); Due Wednesday Sept 17 2025 at 12pm (noon).

The exam is “open book” but you are not permitted to discuss any aspect of this exam with anyone else. Questions of clarification should be addressed to `ignat@uchicago.edu`, `jingshuw@uchicago.edu`, `schein@uchicago.edu`, and `mcpeek@uchicago.edu`.

If you choose to use tools such as ChatGPT, we expect you to do so honorably and responsibly. Do not ask ChatGPT for the answers to the questions directly. If we find *any* evidence of copied-and-pasted responses, you will fail the prelim. You will be expected to explain in detail to the oral examiners any bit of code, math, or text you submit, so do not submit anything you do not understand.

Finally, remember to pace yourself throughout the exam. Take regular breaks, and ensure you eat and sleep. If time becomes limited, focus on delivering complete answers, even if it means leaving some sections unfinished, rather than offering incomplete responses for many questions. We want you to succeed. Good luck and try to have some fun!

## 1 Data

The data set for this exam is taken from the paper

*Effects of disliked music on psychophysiology*, Merrill et al, Scientific Reports, 2023, available at <https://www.nature.com/articles/s41598-023-46963-7#MOESM1>

Two data files are provided: (i) an Excel workbook (`merrill_data_part1.xlsx`) detailing participant selection and characteristics (variable definitions are in Tables 1-2), and (ii) a CSV file (`merrill_data_part2.csv`) containing the Session 2 main-experiment data (variable definitions in Table 3). Two additional data files (`test_data_age.csv`, `test_data_ratings.csv`) will be used for Question 6.

## 2 Questions

Your report should include code for all plots and results that you present. The report should be *concise, well-organized, and easy to read*. For example, if you examine plots across a long list of different subsets of the data, you may choose to show only a small selection of representative plots.

1. Please read the original paper, state the primary scientific question and summarize the design (screening → Session 1 matching → Session 2 main experiment), including how participants, songs, and real-time ratings/physiology were obtained. Also, summarize the main trends observed or conclusions drawn by the authors. Highlight some of the main statistical issues, as you see them, that might be present in this data set or in the analysis.
2. Consider the `online_survey` sheet in the Excel file, which records covariates `age` and `gender` for all participants. Here `gender` is coded as taking values in  $\{1, 2, 3\}$ .
  - (a) Use the data in this sheet to provide statistical evidence for or against the claim that participants with `gender` = 1 dislike the music of the artist Helene Fischer more strongly than participants with `gender` = 2.
  - (b) Would any other information provided in this paper enable you to strengthen your analysis with respect to this question? If so, how? (You do not need to conduct the analysis, just describe what you would do.)
3. The sheet `session1` of the Excel file records the rating assigned to each song by each participant in Session 1.
  - (a) Propose a metric for ranking songs by popularity that can be computed from the ratings in this spreadsheet. (Example: For each song, report the lowest rating it received.) Compute this metric for each song in the spreadsheet. Explain your rationale for choosing this metric. In the spreadsheet, some entries are colored red, green or blue. Do these entries require special treatment in your proposed metric? Why or why not?
  - (b) Compute the average song rating (averaged over all songs and participants) and provide a 95% confidence interval thereof. What are your assumptions? What estimand is your confidence interval targeting?

- (c) Suppose the researchers randomly pick a new song from Spotify’s Top50-Germany playlist (i.e., a song that is not in the current spreadsheet) and that all participants of session 1 rate this song. Let  $\bar{z}$  be the average rating of this new song across all participants. Consider a game against the other PhD students in your cohort which works as follows: Each of you suggests an interval  $\mathcal{I}$ . The smallest interval (smallest length  $|\mathcal{I}|$ ) that contains  $\bar{z}$  (i.e.,  $\bar{z} \in \mathcal{I}$ ) is the winner. What is your proposed interval? Explain your statistical rationale.
4. In the CSV from the main experiment, eight physiological measures are recorded: {EMGCS, EMGZM, ECGHR, EDA\_phasic, EMGLLSAN, PPGHR, RRbpm, temperature}. For each measure, compute for each individual the mean within each condition (*disliked, neutral; labeled “indifferent” in the file*) so that each individual has two values per measure (one per condition).
- (a) Using these paired summaries, assess whether any of the eight measures differ between disliked and neutral music. Present your results using nice and clear figures and/or tables.
- (b) Participants heard six pieces (3 disliked, 3 neutral), each  $\sim 3$  minutes, in sequence. The goal is to estimate the effect of hearing disliked music, but ordering of songs may bias participants’ judgments. In the paper, the six pieces were arranged into two blocks (three disliked; three neutral); *block order* was counterbalanced (half random participants get one order, and others get the other order) across participants, and song order within blocks was randomized. Is this strategy sufficient to remove order-induced bias when estimating the effect of disliked music? Write down a statistical framework to quantitatively explain why counterbalancing may/may not work and what assumptions are needed.
5. The paper fits a linear mixed model on z-transformed physiology with fixed effects of *time window, condition, rating* and a *participant* random intercept (no random slopes). Is this a reasonable modeling strategy? Is any critical information missing from the provided CSV that would affect modeling or inference? Discuss briefly.
6. Participants were asked to rate 19 musical genres (e.g., “rap”, “funk”, “Latin”, “Volksmusik”) on a 7-star scale. This question asks you to analyze cluster structure in ratings of musical genre and explore how/whether that cluster structure relates to age.

Consider the following mixture model:

$$\begin{aligned}
&\text{for participant } i = 1, \dots, n: \\
&\quad z_i \sim \text{Categorical}(\pi_1, \dots, \pi_K) \quad \# \text{ sample cluster assignment} \\
&\quad y_i^{\text{age}} \sim \text{Poisson}(\lambda_{z_i}) \quad \# \text{ sample age, given cluster} \\
&\quad \text{for genre } j = 1, \dots, m: \\
&\quad \quad y_{i,j}^{\text{rating}} \sim \text{Binomial}(7, \phi_{z_i,j}) \quad \# \text{ sample rating, given cluster}
\end{aligned}$$

The following terms are defined:

- There are  $n = 529$  unique participants in the data set, each indexed by  $i$ .
- Each participant rates  $m = 19$  musical genres, each indexed by  $j$ .
- $y_i^{\text{age}} \in \mathbb{N}$  is the age of participant  $i$ .
- $y_{i,j}^{\text{rating}} \in \{0, 1, \dots, 7\}$  is the rating  $i$  gives to genre  $j$ .
- $z_i \in \{1, \dots, K\}$  is the assigned mixture component (or “cluster”) of participant  $i$ .
- $\pi_k \in (0, 1)$  is the frequency of cluster  $k$  where  $\sum_{k=1}^K \pi_k = 1$ .
- $\lambda_k > 0$  is the mean age of participants in cluster  $k$ .
- $\phi_{k,j} \in (0, 1)$  represents the popularity of genre  $j$  in cluster  $k$ .

For convenience, we will also assume the following non-informative prior distributions:

$$\begin{aligned}
&(\pi_1, \dots, \pi_K) \sim \text{Dirichlet}(\alpha_0, \dots, \alpha_0) \\
&\text{for cluster } k = 1, \dots, K: \\
&\quad \lambda_k \sim \text{Gamma}(s_0, r_0) \quad \# \text{ shape and rate parameterization} \\
&\quad \text{for genre } j = 1, \dots, m: \\
&\quad \quad \phi_{k,j} \sim \text{Beta}(a_0, b_0)
\end{aligned}$$

where the hyperparameters are set to  $\alpha_0 = 1.15$ ,  $s_0 = 1.15$ ,  $r_0 = 1/30$ ,  $a_0 = 1.15$ ,  $b_0 = 1.15$ .

- (a) Derive an expectation-maximization (EM) algorithm to perform maximum-a-posteriori (MAP) estimation of the  $\pi_k$ ,  $\lambda_k$ , and  $\phi_{k,j}$  parameters. You should give a clear description of the full algorithm and provide a complete derivation which clearly explains and justifies each step.
- (b) Implement the EM algorithm in a programming language of your choice. Your implementation should be numerically stable, reasonably efficient, and correct. It should also include comments that explain the code.
- (c) Fit the model to the data using your implementation of EM.
- The data of ratings and age can be found in `merrill_data_part1.xlsx`, where the columns for musical genres all begin with `m_` (e.g., `m_funk`), and the column for age is `age`.
  - We suggest setting  $K = 5$ . You may use a different value as long as you explain why.
  - There are different ways to initialize the parameters, but we suggest sampling them from the prior.
  - EM is an iterative algorithm; make sure you run it until convergence.
- (d) Perform an exploratory analysis of the inferred latent structure. What musical genres cluster together? Which age groups prefer which clusters? Your analysis should include high-quality visualizations and show patterns in the data that the model seems to uncover. You may also choose to incorporate other covariates or information from the broader dataset. (This question is intentionally left open-ended; use your creativity.)
- (e) We have included a test set of  $n^{\text{test}} = 100$  participants whose ratings can be found in `test_data_ratings.csv`. Use the parameters learned by EM in the last question—i.e.,  $\hat{\lambda}, \hat{\Phi}, \hat{\pi}$ —to predict the age of the test participants given their ratings. More specifically, for each test participant, compute the posterior predictive expectation of their age  $y_i^{\text{age}}$  given their ratings  $\mathbf{y}_i^{\text{rating}} \equiv (y_{i,1}^{\text{rating}}, \dots, y_{i,m}^{\text{rating}})$  and the learned parameters:

$$\mathbb{E}[y_i^{\text{age}} \mid \mathbf{y}_i^{\text{rating}}] = \sum_{y=0}^{\infty} y P(y_i^{\text{age}} = y \mid \mathbf{y}_i^{\text{rating}}, \hat{\lambda}, \hat{\Phi}, \hat{\pi})$$

Design a procedure to compute this and explain it clearly. Then implement it, run it on the test set, and compare the model's predictions to the ground truth using an appropriate set of performance metrics and visualizations. The ground truth age of the test participants can be found in `test_data_age.csv`.

- (f) The model above makes very specific parametric assumptions which may or may not be warranted and/or correct. Suggest three distinct ways in which the model might be improved. An example of such a change might be: “Change the Poisson assumption to a negative binomial assumption to model an overdispersed distribution over age.” Argue persuasively for why those changes might be justified, citing your knowledge of the data and your analysis of the current model's fit. (Again this is very open-ended; use your creativity.)

Table 1: Online survey variables and definitions. Prefix **sN\_** denotes song  $N \in \{1, 2, 3\}$ .

| Variable                                                                                                                                                                                 | Description                                                        | Type / Scale  | Notes                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------|---------------------------------------|
| Participant                                                                                                                                                                              | Anonymous participant ID                                           | integer       |                                       |
| age                                                                                                                                                                                      | Age in years                                                       | numeric       |                                       |
| gender                                                                                                                                                                                   | Self-reported gender                                               | categorical   |                                       |
| sN.title                                                                                                                                                                                 | Title of disliked piece $N$ named in the survey                    | text          |                                       |
| sN.artist                                                                                                                                                                                | Artist of disliked piece $N$                                       | text          |                                       |
| sN.grade                                                                                                                                                                                 | Overall dislike rating for piece $N$                               | 5-point scale | Used in screening; threshold $\geq 4$ |
| sN.boring                                                                                                                                                                                | "I find it too boring."                                            | 5-point scale | Filler item                           |
| sN.displ                                                                                                                                                                                 | "It evoked unpleasant feelings in me."                             | 5-point scale | High-arousal set                      |
| sN.no_impact                                                                                                                                                                             | "It has no impact on my mood."                                     | 5-point scale | Low-arousal set                       |
| sN.lyrics                                                                                                                                                                                | "I find the lyrics bad."                                           | 5-point scale | Filler item                           |
| sN.stress                                                                                                                                                                                | "I feel stressed."                                                 | 5-point scale | High-arousal set                      |
| sN.exper                                                                                                                                                                                 | Other/filler item (label "exper," not explicitly defined in paper) | 5-point scale |                                       |
| sN.mood                                                                                                                                                                                  | "It puts me in a bad mood."                                        | 5-point scale | High-arousal set                      |
| sN.sad                                                                                                                                                                                   | "It makes me sad."                                                 | 5-point scale | Low-arousal set                       |
| sN.touched                                                                                                                                                                               | "I am not moved by it."                                            | 5-point scale | Low-arousal set                       |
| sN.expr                                                                                                                                                                                  | "It has too little emotional expression."                          | 5-point scale | Filler item                           |
| sN.aggre                                                                                                                                                                                 | "It makes me aggressive."                                          | 5-point scale | High-arousal set                      |
| sN.mus                                                                                                                                                                                   | "I find it musically bad."                                         | 5-point scale | Filler item                           |
| sN.body_displ                                                                                                                                                                            | "It makes me feel physically uncomfortable."                       | 5-point scale | High-arousal set                      |
| sN.memory                                                                                                                                                                                | "I associate negative memories with it."                           | 5-point scale | Exclusion reason in screening         |
| m.blues, m.jazz, m.funk, m.soul,<br>m.reggae, m.techno, m.edm,<br>m.house, m.rap, m.kultur,<br>m.latin, m.weltm, m.klassik,<br>m.volks, m.schlager, m.country,<br>m.pop, m.rock, m.metal | Genre preference                                                   | 7-point scale | 1= strongly dislike, 7=like very much |

Table 2: Column names and descriptions (Session 2 trait/demographics file).

| Column                                                                                                                                                                                                                                                                                                                                                                | Description                                                            | Instrument             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------|
| Participant                                                                                                                                                                                                                                                                                                                                                           | Participant ID                                                         | —                      |
| T_Ang                                                                                                                                                                                                                                                                                                                                                                 | Trait Anger (total)                                                    | STAXI-2                |
| T_Ang_T                                                                                                                                                                                                                                                                                                                                                               | Trait Anger—Temperament subscale                                       | STAXI-2                |
| T_Ang_R                                                                                                                                                                                                                                                                                                                                                               | Trait Anger—Reaction subscale                                          | STAXI-2                |
| AX_O                                                                                                                                                                                                                                                                                                                                                                  | Anger Expression Out                                                   | STAXI-2                |
| AX_I                                                                                                                                                                                                                                                                                                                                                                  | Anger Expression In                                                    | STAXI-2                |
| AC                                                                                                                                                                                                                                                                                                                                                                    | Anger Control (total)                                                  | STAXI-2                |
| AC_O                                                                                                                                                                                                                                                                                                                                                                  | Anger Control Out                                                      | STAXI-2                |
| AC_I                                                                                                                                                                                                                                                                                                                                                                  | Anger Control In                                                       | STAXI-2                |
| P_Extra                                                                                                                                                                                                                                                                                                                                                               | Extraversion (Big Five)                                                | BFI-2                  |
| P_Agree                                                                                                                                                                                                                                                                                                                                                               | Agreeableness                                                          | BFI-2                  |
| P_Consci                                                                                                                                                                                                                                                                                                                                                              | Conscientiousness                                                      | BFI-2                  |
| P_neg_emo                                                                                                                                                                                                                                                                                                                                                             | Negative Emotionality                                                  | BFI-2                  |
| P_open_mind                                                                                                                                                                                                                                                                                                                                                           | Open-Mindedness                                                        | BFI-2                  |
| PS_socia                                                                                                                                                                                                                                                                                                                                                              | Sociability (Extraversion facet)                                       | BFI-2 facets           |
| PS_assert                                                                                                                                                                                                                                                                                                                                                             | Assertiveness (Extraversion facet)                                     | BFI-2 facets           |
| PS_energ                                                                                                                                                                                                                                                                                                                                                              | Energy Level (Extraversion facet)                                      | BFI-2 facets           |
| PS_compa                                                                                                                                                                                                                                                                                                                                                              | Compassion (Agreeableness facet)                                       | BFI-2 facets           |
| PS_respec                                                                                                                                                                                                                                                                                                                                                             | Respectfulness (Agreeableness facet)                                   | BFI-2 facets           |
| PS_trust                                                                                                                                                                                                                                                                                                                                                              | Trust (Agreeableness facet)                                            | BFI-2 facets           |
| PS_organ                                                                                                                                                                                                                                                                                                                                                              | Organization (Conscientiousness facet)                                 | BFI-2 facets           |
| PS_produc                                                                                                                                                                                                                                                                                                                                                             | Productiveness (Conscientiousness facet)                               | BFI-2 facets           |
| PS_respo                                                                                                                                                                                                                                                                                                                                                              | Responsibility (Conscientiousness facet)                               | BFI-2 facets           |
| PS_anxie                                                                                                                                                                                                                                                                                                                                                              | Anxiety (Neg. Emotionality facet)                                      | BFI-2 facets           |
| PS_depres                                                                                                                                                                                                                                                                                                                                                             | Depression (Neg. Emotionality facet)                                   | BFI-2 facets           |
| PS_volati                                                                                                                                                                                                                                                                                                                                                             | Emotional Volatility (Neg. Emotionality facet)                         | BFI-2 facets           |
| PS_aesth                                                                                                                                                                                                                                                                                                                                                              | Aesthetic Sensitivity (Openness facet)                                 | BFI-2 facets           |
| PS_curios                                                                                                                                                                                                                                                                                                                                                             | Intellectual Curiosity (Openness facet)                                | BFI-2 facets           |
| PS_imagin                                                                                                                                                                                                                                                                                                                                                             | Creative Imagination (Openness facet)                                  | BFI-2 facets           |
| NCCS                                                                                                                                                                                                                                                                                                                                                                  | Need for Cognitive Closure (total)                                     | 16-NCCS                |
| GMSI_active                                                                                                                                                                                                                                                                                                                                                           | Active Engagement with Music                                           | Gold-MSI               |
| GMSI_perce                                                                                                                                                                                                                                                                                                                                                            | Perceptual Abilities                                                   | Gold-MSI               |
| GMSI_train                                                                                                                                                                                                                                                                                                                                                            | Musical Training                                                       | Gold-MSI               |
| GMSI_sing                                                                                                                                                                                                                                                                                                                                                             | Singing Abilities                                                      | Gold-MSI               |
| GMSI_emo                                                                                                                                                                                                                                                                                                                                                              | Musical Emotions                                                       | Gold-MSI               |
| GMSI_gener                                                                                                                                                                                                                                                                                                                                                            | General Musical Sophistication                                         | Gold-MSI               |
| GMSI_instr                                                                                                                                                                                                                                                                                                                                                            | Instrumental Abilities                                                 | Gold-MSI               |
| PSRS_tot                                                                                                                                                                                                                                                                                                                                                              | Perceived Stress Reactivity (total)                                    | PSRS                   |
| PSRS_AR                                                                                                                                                                                                                                                                                                                                                               | Anticipatory Reactivity                                                | PSRS                   |
| PSRS_RM                                                                                                                                                                                                                                                                                                                                                               | Reactivity to Performance/Math tasks                                   | PSRS                   |
| PSRS_RsK                                                                                                                                                                                                                                                                                                                                                              | Reactivity to Social Conflicts                                         | PSRS                   |
| PSRS_RA                                                                                                                                                                                                                                                                                                                                                               | Reactivity to Failure/Achievement                                      | PSRS                   |
| PSRS_RsB                                                                                                                                                                                                                                                                                                                                                              | Reactivity to Work Overload/Burden                                     | PSRS                   |
| school.degree                                                                                                                                                                                                                                                                                                                                                         | Highest school degree completed                                        | Demographics           |
| education                                                                                                                                                                                                                                                                                                                                                             | Highest educational qualification (e.g., BA, MA, PhD)                  | Demographics           |
| income                                                                                                                                                                                                                                                                                                                                                                | Income category                                                        | Demographics           |
| profession01                                                                                                                                                                                                                                                                                                                                                          | Profession / occupation (primary code)                                 | Demographics           |
| profession02a                                                                                                                                                                                                                                                                                                                                                         | Profession (secondary code A)                                          | Demographics           |
| profession02b                                                                                                                                                                                                                                                                                                                                                         | Profession (secondary code B)                                          | Demographics           |
| T_Ang_m, T_Ang_T_m, T_Ang_R_m,<br>AX_O_m, AX_I_m, AC_m, AC_O_m,<br>AC_I_m, P_Extra_m, P_Agree_m,<br>P_Consci_m, P_neg_emo_m,<br>P_open_mind_m, PS_socia_m,<br>PS_assert_m, PS_energ_m,<br>PS_compa_m, PS_respec_m,<br>PS_trust_m, PS_organ_m,<br>PS_produc_m, PS_respo_m,<br>PS_anxie_m, PS_depres_m,<br>PS_volati_m, PS_aesth_m,<br>PS_curios_m, PS_imagin_m, NCCS_m | Mean-centered / z-standardized versions of the corresponding variables | (same as source above) |

Table 3: Part 2 data column definitions

| Column                  | Description                                                                                                          | Common values / units                                                                             |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>value</b>            | Numeric measurement for the specified physiology channel within the specified 3 s time window around a button press. | e.g., BPM, $\mu$ S, $^{\circ}$ C, EMG envelope (a.u.)                                             |
| <b>phys id</b>          | Physiological channel/signal.<br>Participant identifier (anonymous numeric code).                                    | EMGCS, EMGZM, EMGLLSAN, ECGHR, PPGHR, EDA_phasic, RRbpm, temperature<br>e.g., 1065                |
| <b>condition rating</b> | Music condition at the time of the event.<br>Real-time button label at the event (valence scale).                    | disliked (self-selected), indifferent (neutral)<br>pleasant, neutral, unpleasant, very unpleasant |
| <b>tw</b>               | 3 s time window relative to the button press within a 12 s epoch.                                                    | pre (−3–0 s), 1st (0–3 s), 2nd (3–6 s), 3rd (6–9 s)                                               |